

EXTENSION 2

Stage 6

Calculus (Ext2), C1 Integration (Ext2)

Substitution and Harder Integration

Teacher: Nardene Dodd

Exam Equivalent Time: 118.5 minutes (based on allocation of 1.5 minutes per mark)



Questions

1. Calculus, EXT2 C2 2020 SPEC2 11 MC

With a suitable substitution $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2(x)}{\sec^2(x) - 3\tan(x) + 1} \, dx$ can be expressed as

- A. $\int_1^{\frac{1}{\sqrt{3}}} \left(\frac{1}{u-1} - \frac{1}{u-2} \right) \, du$
- B. $\int_1^{\sqrt{3}} \left(\frac{1}{u-2} - \frac{1}{u-1} \right) \, du$
- C. $\int_1^{\sqrt{3}} \left(\frac{1}{u-1} - \frac{1}{u-2} \right) \, du$
- D. $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{1}{3(u-1)} - \frac{1}{3(u+2)} \right) \, du$

2. Calculus, EXT2 C1 2022 HSC 4 MC

Of the following expressions, which one need NOT contain a term involving a logarithm in its anti-derivative?

- A. $\frac{x+2}{x^2+4x+5}$
- B. $\frac{x+2}{x^2-4x-5}$
- C. $\frac{x-1}{x^3-x^2+x-1}$
- D. $\frac{x+1}{x^3-x^2+x-1}$

3. Calculus, EXT2 C1 2017 HSC 7 MC

It is given that $f(x)$ is a non-zero even function and $g(x)$ is a non-zero odd function.

Which expression is equal to $\int_{-a}^a f(x) + g(x) \, dx$?

- A. $2\int_0^a f(x) \, dx$
- B. $2\int_0^a g(x) \, dx$
- C. $\int_{-a}^a g(x) \, dx$
- D. $2\int_0^a f(x) + g(x) \, dx$

4. Calculus, EXT2 C1 2022 HSC 5 MC

If $\int_a^x f(t)dt = g(x)$, which of the following is a primitive of $f(x)g(x)$?

- A. $\frac{1}{2}[f(x)]^2$
- B. $\frac{1}{2}[f'(x)]^2$
- C. $\frac{1}{2}[g(x)]^2$
- D. $\frac{1}{2}[g'(x)]^2$

5. Calculus, EXT2 C1 2020 HSC 10 MC

Which of the following is equal to $\int_0^{2a} f(x) \, dx$?

- A. $\int_0^a f(x) - f(2a-x) \, dx$
- B. $\int_0^a f(x) + f(2a-x) \, dx$
- C. $2\int_0^a f(x-a) \, dx$
- D. $\int_0^a \frac{1}{2}f(2x) \, dx$

6. Calculus, EXT2 C1 2012 HSC 10 MC

Without evaluating the integrals, which one of the following integrals is greater than zero?

- A. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x}{2 + \cos x} dx$
- B. $\int_{-\pi}^{\pi} x^3 \sin x dx$
- C. $\int_{-1}^1 (e^{-x^2} - 1) dx$
- D. $\int_{-2}^2 \tan^{-1}(x^3) dx$

7. Calculus, EXT2 C1 2014 HSC 10 MC

Which integral is necessarily equal to $\int_{-a}^a f(x) dx$?

- A. $\int_0^a f(x) - f(-x) dx$
- B. $\int_0^a f(x) - f(a - x) dx$
- C. $\int_0^a f(x - a) + f(-x) dx$
- D. $\int_0^a f(x - a) + f(a - x) dx$

8. Calculus, EXT2 C1 2006 HSC 1a

Find $\int \frac{x}{\sqrt{9 - 4x^2}} dx$. (2 marks)

9. Calculus, EXT2 C1 2011 HSC 1b

Evaluate $\int_0^3 x\sqrt{x+1} dx$. (3 marks)

10. Calculus, EXT2 C1 2005 HSC 1a

Find $\int \frac{\cos \theta}{\sin^5 \theta} d\theta$ (2 marks)

11. Calculus, EXT2 C1 2019 SPEC1-N 4

Evaluate $\int_{e^3}^{e^4} \frac{1}{x \log_e(x)} dx$. (3 marks)

12. Calculus, EXT2 C1 2020 SPEC1 2

Evaluate $\int_{-1}^0 \frac{1+x}{\sqrt{1-x}} dx$. (3 marks)

13. Calculus, EXT2 C1 2022 HSC 11f

Using the substitution $t = \tan \frac{x}{2}$, find

$\int \frac{dx}{1 + \cos x - \sin x}$ (3 marks)

14. Calculus, EXT2 C1 2009 HSC 1a

Find $\int \frac{\ln x}{x} dx$. (2 marks)

15. Calculus, EXT2 C1 2010 HSC 1a

Find $\int \frac{x}{\sqrt{1+3x^2}} dx$. (2 marks)

16. Calculus, EXT2 C1 2010 HSC 1b

Evaluate $\int_0^{\frac{\pi}{4}} \tan x dx$. (3 marks)

17. Calculus, EXT2 C1 2012 HSC 11e

Evaluate $\int_0^1 \frac{e^{2x}}{e^{2x} + 1} dx$. (3 marks)

18. Calculus, EXT2 C1 2017 HSC 11f

Using the substitution $x = \sin^2 \theta$, or otherwise, evaluate $\int_0^{\frac{1}{2}} \sqrt{\frac{x}{1-x}} dx$. (3 marks)

19. Calculus, EXT2 C1 2019 HSC 15a

i. Show that

$$\int_{-a}^a \frac{f(x)}{f(x) + f(-x)} dx = \int_{-a}^a \frac{f(-x)}{f(x) + f(-x)} dx. \quad (2 \text{ marks})$$

ii. Hence, or otherwise, evaluate

$$\int_{-1}^1 \frac{e^x}{e^x + e^{-x}} dx. \quad (2 \text{ marks})$$

20. Calculus, EXT2 C1 2021 HSC 13b

Use an appropriate substitution to evaluate $\int_{\sqrt{10}}^{\sqrt{13}} x^3 \sqrt{x^2 - 9} \, dx$. (3 marks)

21. Calculus, EXT2 C1 2022 HSC 15c

Using the substitution $x = \tan^2 \theta$, evaluate

$$\int_0^1 \sin^{-1} \sqrt{\frac{x}{1+x}} \, dx \quad (4 \text{ marks})$$

22. Calculus, EXT2 C1 2023 HSC 13a

Find $\int \frac{1-x}{\sqrt{5-4x-x^2}} \, dx$. (3 marks)

23. Calculus, EXT2 C1 2007 HSC 1d

Evaluate $\int_0^{\frac{3}{4}} \frac{x}{\sqrt{1-x}} \, dx$. (4 marks)

24. Calculus, EXT2 C1 2007 HSC 5c

- Write $(x-1)(5-x)$ in the form $b^2 - (x-a)^2$, where a and b are real numbers. (1 mark)
- Using the values of a and b found in part (i) and making the substitution $x-a = b \sin \theta$, or otherwise, evaluate

$$\int_1^5 \sqrt{(x-1)(5-x)} \, dx. \quad (2 \text{ marks})$$

25. Calculus, EXT2 C1 2009 HSC 1e

Evaluate $\int_1^{\sqrt{3}} \frac{1}{x^2 \sqrt{1+x^2}} \, dx$. (4 marks)

26. Calculus, EXT2 C1 2010 HSC 1e

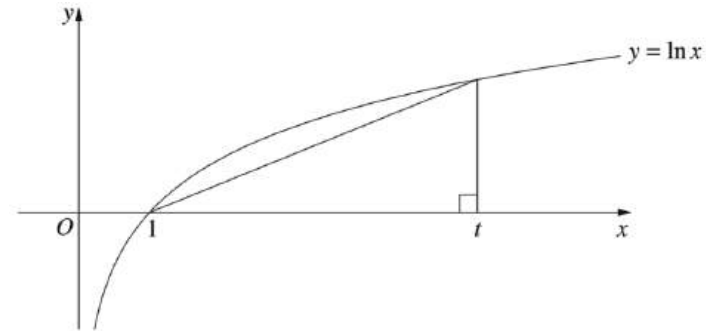
Find $\int \frac{dx}{1+\sqrt{x}}$. (3 marks)

27. Calculus, EXT2 C1 2013 HSC 11d

Evaluate $\int_0^1 x^3 \sqrt{1-x^2} \, dx$. (3 marks)

28. Calculus, EXT2 C1 2013 HSC 14a

The diagram shows the graph $y = \ln x$.



By comparing relevant areas in the diagram, or otherwise, show that

$$\ln t > 2 \left(\frac{t-1}{t+1} \right), \text{ for } t > 1. \quad (3 \text{ marks})$$

29. Calculus, EXT2 C1 2011 HSC 7b

Let $I = \int_1^3 \frac{\cos^2(\frac{\pi}{8}x)}{x(4-x)} \, dx$.

- Use the substitution $u = 4-x$ to show that

$$I = \int_1^3 \frac{\sin^2(\frac{\pi}{8}u)}{u(4-u)} \, du. \quad (2 \text{ marks})$$

- Hence, find the value of I . (3 marks)

30. Calculus, EXT2 C1 2007 HSC 8a

- Using a suitable substitution, show that

$$\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx. \quad (1 \text{ mark})$$

- A function $f(x)$ has the property that $f(x) + f(a-x) = f(a)$.

Using part (i), or otherwise, show that

$$\int_0^a f(x) \, dx = \frac{a}{2} f(a). \quad (2 \text{ marks})$$

31. Calculus, EXT2 C1 2022 SPEC2 7 MC

Using the substitution $u = 1 + e^x$, $\int_0^{\log_e 2} \frac{1}{1 + e^x} dx$ can be expressed as

- A. $\int_0^{\log_e 2} \left(\frac{1}{u-1} - \frac{1}{u} \right) du$
 B. $\int_2^3 \left(\frac{1}{u} - \frac{1}{u-1} \right) du$
 C. $\int_1^3 \left(\frac{1}{u} - \frac{1}{u-1} \right) du$
 D. $\int_2^3 \left(\frac{1}{u-1} - \frac{1}{u} \right) du$

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Worked Solutions

1. Calculus, EXT2 C2 2020 SPEC2 11 MC

Let $u = \tan(x)$

$$\frac{du}{dx} = \sec^2(x) \Rightarrow du = \sec^2(x) dx$$

When $x = \frac{\pi}{3}$, $u = \sqrt{3}$

$$x = \frac{\pi}{4}, u = 1$$

$$\begin{aligned} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2(x)}{\sec^2(x) - 3\tan(x) + 1} dx &= \int_1^{\sqrt{3}} \frac{1}{u^2 + 1 - 3u + 1} du \\ &= \int_1^{\sqrt{3}} \frac{1}{u^2 - 3u + 2} du \\ &= \int_1^{\sqrt{3}} \frac{1}{(u-2)(u-1)} du \\ &= \int_1^{\sqrt{3}} \frac{1}{u-2} - \frac{1}{u-1} du \end{aligned}$$

$\Rightarrow B$

2. Calculus, EXT2 C1 2022 HSC 4 MC

Consider the denominator of C :

$$\begin{aligned} x^3 - x^2 + x - 1 &= x^2(x-1) + (x-1) \\ &= (x^2 + 1)(x-1) \end{aligned}$$

$$\frac{x-1}{x^3 - x^2 + x - 1} = \frac{x-1}{(x^2 + 1)(x-1)} = \frac{1}{x^2 + 1}$$

$$\int \frac{1}{x^2 + 1} dx = \tan^{-1}(x) + c$$

$\Rightarrow C$

3. Calculus, EXT2 C1 2017 HSC 7 MC

$$\begin{aligned} \int_{-a}^a f(x) + g(x) dx &= \int_{-a}^a f(x) dx + \int_{-a}^a g(x) dx \\ &= 2 \int_0^a f(x) dx + 0 \end{aligned}$$

$\Rightarrow A$

4. Calculus, EXT2 C1 2022 HSC 5 MC

$$\begin{aligned}\int_a^x f(t) \, dt &= g(x) \\ \frac{d}{dx} \left[\int_a^x f(t) \, dt \right] &= g'(x) \\ f(x) &= g'(x) \\ f(x) \cdot g(x) &= g'(x)g(x) \\ \int f(x) \cdot g(x) \, dx &= \int g'(x)g(x) \, dx \\ &= \frac{1}{2}[g(x)]^2 + C \\ &\Rightarrow C\end{aligned}$$

COMMENT: Simple examples can illustrate the logic of line 3 of the worked solution.

5. Calculus, EXT2 C1 2020 HSC 10 MC

$$\begin{aligned}\int_0^{2a} f(x) \, dx - \int_0^a f(x) \, dx + \int_a^{2a} f(x) \, dx \\ \text{Let } x = 2a - u \Rightarrow u = 2a - x \\ \frac{du}{dx} = -1 \Rightarrow du = -dx\end{aligned}$$

♦ Mean mark 54%.

$$\begin{aligned}\text{When } x = a, \quad u &= a \\ x = 2a, \quad u &= 0\end{aligned}$$

$$\begin{aligned}\int_0^{2a} f(x) \, dx &= \int_0^a f(x) \, dx - \int_a^0 f(2a - u) \, du \\ \text{Use substitution for } \int_a^0 f(2a - u) \, du\end{aligned}$$

$$\text{Let } 2a - u = 2a - x \Rightarrow x = u$$

$$\frac{dx}{du} = 1 \Rightarrow du = dx$$

$$\begin{aligned}\text{When } u = a, \quad x &= a \\ u = 0, \quad x &= 0\end{aligned}$$

$$\begin{aligned}\therefore \int_0^{2a} f(x) \, dx &= \int_0^a f(x) \, dx - \int_a^0 f(2a - x) \, dx \\ &= \int_0^a f(x) \, dx + \int_0^a f(2a - x) \, dx \\ &= \int_0^a f(x) + f(2a - x) \, dx\end{aligned}$$

$$\Rightarrow B$$

6. Calculus, EXT2 C1 2012 HSC 10 MC

Consider (A) and (D)

$$f(x) = -f(-x) \Rightarrow \text{ODD functions where}$$

$$\int_{-a}^a f(x) \, dx = 0$$

Consider (C)

$$e^{-x^2} < 1 \text{ for all } x \Rightarrow e^{-x^2} - 1 < 0$$

$\therefore$ Its graph is below the x -axis and any integral will be negative

Consider (B)

(B) is an even function where,

$$x^3 \sin x \geq 0 \text{ for } -\pi \leq x \leq \pi \\ \Rightarrow B$$

7. Calculus, EXT2 C1 2014 HSC 10 MC

$$\int_{-a}^a f(x) \, dx = \int_0^a f(x) \, dx + \int_{-a}^0 f(x) \, dx$$

$$\text{Consider } \int_0^a f(x) \, dx$$

$$\text{Let } u = a - x \Rightarrow x = a - u \text{ and } dx = -du$$

$$\int_0^a f(x) \, dx = \int_0^a f(a - u) - du$$

$$= \int_a^0 f(a - u) \, du$$

$$= \int_0^a f(a - x) \, dx$$

$$\text{Consider } \int_{-a}^0 f(x) \, dx$$

$$\text{Let } u = x + a \Rightarrow x = u - a \text{ and } dx = du$$

$$\int_{-a}^0 f(x) \, dx = \int_0^a f(u - a) \, du$$

$$= \int_0^a f(x - a) \, dx$$

$$\therefore \int_{-a}^a f(x) \, dx = \int_0^a f(x - a) + f(a - x) \, dx$$

$$\Rightarrow D$$

♦♦ Mean mark 29%.

8. Calculus, EXT2 C1 2006 HSC 1a

$$\int \frac{x}{\sqrt{9-4x^2}} \, dx = -\frac{1}{8} \int \frac{-8x}{\sqrt{9-4x^2}} \, dx \\ = -\frac{1}{4} \sqrt{9-4x^2} + c$$

9. Calculus, EXT2 C1 2011 HSC 1b

$$\text{Let } u = 1 + x, \quad du = dx$$

$$\text{When } x = 0, \quad u = 1$$

$$x = 3, \quad u = 4$$

$$\int_0^3 x\sqrt{1+x} \, dx = \int_1^4 (u-1)\sqrt{u} \, du \\ = \int_1^4 \left(u^{\frac{3}{2}} - u^{\frac{1}{2}}\right) \, du \\ = \left[\frac{2}{5}u^{\frac{5}{2}} - \frac{2}{3}u^{\frac{3}{2}}\right]_1^4 \\ = \left(\frac{2}{5} \times 32 - \frac{2}{3} \times 8\right) - \left(\frac{2}{5} - \frac{2}{3}\right) \\ = \frac{116}{15}$$

10. Calculus, EXT2 C1 2005 HSC 1a

$$\text{Let } u = \sin \theta$$

$$\frac{du}{d\theta} = \cos \theta \Rightarrow du = \cos \theta \, d\theta$$

$$\int \frac{\cos \theta}{\sin^5 \theta} \, d\theta = \int u^{-5} \, du \\ = \frac{-1}{4} u^{-4} + C \\ = \frac{-1}{4 \sin^4 \theta} + C$$

11. Calculus, EXT2 C1 2019 SPEC1-N 4

Let $u = \log_e x$

$$\frac{du}{dx} = \frac{1}{x} \Rightarrow du = \frac{1}{x} dx$$

When $x = e^4 \Rightarrow u = 4$ When $x = e^3 \Rightarrow u = 3$

$$\begin{aligned} \int_{e^3}^{e^4} \frac{1}{x \log_e(x)} dx &= \int_3^4 \frac{1}{u} du \\ &= [\log_e u]_3^4 \\ &= \log_e 4 - \log_e 3 \\ &= \log_e \left(\frac{4}{3} \right) \end{aligned}$$

12. Calculus, EXT2 C1 2020 SPEC1 2

Let $u = 1-x \Rightarrow x = 1-u$

$$\frac{du}{dx} = -1 \Rightarrow dx = -du$$

When $x = 0, u = 1$ $x = -1, u = 2$

$$\begin{aligned} \int_{-1}^0 \frac{1+x}{\sqrt{1-x}} dx &= - \int_2^1 \frac{2-u}{\sqrt{u}} du \\ &= \int_1^2 2u^{-\frac{1}{2}} - u^{\frac{1}{2}} du \\ &= \left[4u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}} \right]_1^2 \\ &= 4\sqrt{2} - \frac{4\sqrt{2}}{3} - \left(4 - \frac{2}{3} \right) \\ &= \frac{8\sqrt{2}}{3} - \frac{10}{3} \end{aligned}$$

13. Calculus, EXT2 C1 2022 HSC 11f

$$\text{Let } t = \tan \frac{x}{2}, \cos x = \frac{1-t^2}{1+t^2}, \sin x = \frac{2t}{1+t^2}$$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx \Rightarrow dx = \frac{2 dt}{\sec^2 \frac{x}{2}} = \frac{2}{1+t^2} dt$$

$$\begin{aligned} I &= \int \frac{dx}{1 + \cos x - \sin x} \\ &= \int \frac{1}{1 + \frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\ &= \int \frac{2}{1+t^2 + 1-t^2 - 2t} dt \\ &= \int \frac{1}{1-t} dt \\ &= -\ln|1-t| + C \\ &= -\ln \left| 1 - \tan \left(\frac{x}{2} \right) \right| + C \end{aligned}$$

14. Calculus, EXT2 C1 2009 HSC 1a

$$\text{Let } u = \ln x, \quad du = \frac{1}{x} dx$$

$$\begin{aligned} \int \frac{\ln x}{x} dx &= \int u du \\ &= \frac{1}{2} u^2 + c \\ &= \frac{1}{2} (\ln x)^2 + c \end{aligned}$$

15. Calculus, EXT2 C1 2010 HSC 1a

$$\text{Let } u = 1 + 3x^2, \quad du = 6x dx$$

$$\begin{aligned} \therefore \int \frac{x}{\sqrt{1+3x^2}} dx &= \frac{1}{6} \int u^{-\frac{1}{2}} du \\ &= \frac{1}{6} \times 2 \times \sqrt{u} + c \\ &= \frac{1}{3} \sqrt{1+3x^2} + c \end{aligned}$$

16. Calculus, EXT2 C1 2010 HSC 1b

$$\begin{aligned}
 \int_0^{\frac{\pi}{4}} \tan x \, dx &= \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x} \, dx \\
 &= \left[-\ln \cos x \right]_0^{\frac{\pi}{4}} \\
 &= \left[-\ln \cos \frac{\pi}{4} - (-\ln \cos 0) \right] \\
 &= -\ln \frac{1}{\sqrt{2}} + \ln 1 \\
 &= \ln \sqrt{2} \\
 &= \frac{1}{2} \ln 2
 \end{aligned}$$

17. Calculus, EXT2 C1 2012 HSC 11e

$$\begin{aligned}
 \int_0^1 \frac{e^{2x}}{e^{2x} + 1} \, dx &= \frac{1}{2} \left[\log_e (e^{2x} + 1) \right]_0^1 \\
 &= \frac{1}{2} \left[\log_e (e^2 + 1) - \log_e 2 \right] \\
 &= \frac{1}{2} \log_e \left(\frac{e^2 + 1}{2} \right)
 \end{aligned}$$

18. Calculus, EXT2 C1 2017 HSC 11f

$$x = \sin^2 \theta$$

$$dx = 2 \sin \theta \cos \theta \, d\theta$$

$$\text{If } x = \frac{1}{2}, \sin \theta = \frac{1}{\sqrt{2}}, \theta = \frac{\pi}{4}$$

$$\text{If } x = 0, \theta = 0$$

$$\begin{aligned}
 \int_0^{\frac{1}{2}} \sqrt{\frac{x}{1-x}} \, dx &= \int_0^{\frac{\pi}{4}} \sqrt{\frac{\sin^2 \theta}{1-\sin^2 \theta}} \times 2 \sin \theta \cos \theta \, d\theta \\
 &= \int_0^{\frac{\pi}{4}} \frac{\sin \theta}{\cos \theta} \times 2 \sin \theta \cos \theta \, d\theta \\
 &= \int_0^{\frac{\pi}{4}} 2 \sin^2 \theta \, d\theta \\
 &= \int_0^{\frac{\pi}{4}} 1 - \cos 2\theta \, d\theta \\
 &= \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{4}} \\
 &= \left(\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} \right) - (0 - 0) \\
 &= \frac{\pi}{4} - \frac{1}{2}
 \end{aligned}$$

19. Calculus, EXT2 C1 2019 HSC 15a

i. Let $u = -x \Rightarrow du = -dx$

When $x = a$, $u = -a$

When $x = -a$, $u = a$

$$\begin{aligned}\int_{-a}^a \frac{f(x)}{f(x) + f(-x)} dx &= - \int_a^{-a} \frac{f(-u)}{f(-u) + f(u)} du \\ &= \int_{-a}^a \frac{f(-u)}{f(u) + f(-u)} du \\ &= \int_{-a}^a \frac{f(-x)}{f(x) + f(-x)} dx\end{aligned}$$

ii. Let $I = \int_{-1}^1 \frac{e^x}{e^x + e^{-x}} dx$

$$2I = \int_{-1}^1 \frac{e^x}{e^x + e^{-x}} dx + \int_{-1}^1 \frac{e^{-x}}{e^x + e^{-x}} dx$$

$$2I = \int_{-1}^1 \frac{e^x + e^{-x}}{e^x + e^{-x}} dx$$

$$2I = [x]_{-1}^1$$

$$2I = 2$$

$$\therefore \int_{-1}^1 \frac{e^x}{e^x + e^{-x}} dx = 1$$

20. Calculus, EXT2 C1 2021 HSC 13b

$$\int_{\sqrt{10}}^{\sqrt{13}} x^3 \sqrt{x^2 - 9} dx$$

Let $u = x^2 - 9 \Rightarrow x^2 = u + 9$

$$\frac{du}{dx} = 2x \Rightarrow du = 2x dx$$

If $x = \sqrt{13}$, $u = 4$

If $x = \sqrt{10}$, $u = 1$

$$\begin{aligned}\int_{\sqrt{10}}^{\sqrt{13}} x^3 \sqrt{x^2 - 9} dx &= \int_1^4 (u + 9) \sqrt{u} \cdot \frac{1}{2} du \\ &= \frac{1}{2} \int_1^4 u^{\frac{3}{2}} + 9u^{\frac{1}{2}} du \\ &= \frac{1}{2} \left[\frac{2}{5} u^{\frac{5}{2}} + 9 \cdot \frac{2}{3} u^{\frac{3}{2}} \right]_1^4 \\ &= \frac{1}{2} \left[\left(\frac{2}{5} \cdot 4^{\frac{5}{2}} + 6 \cdot 4^{\frac{3}{2}} \right) - \left(\frac{2}{5} + 6 \right) \right] \\ &= \frac{1}{2} \left(\frac{64}{5} + 48 - \frac{32}{5} \right) \\ &= \frac{136}{5}\end{aligned}$$

21. Calculus, EXT2 C1 2022 HSC 15c

$$x = \tan^2 \theta$$

$$\frac{dx}{d\theta} = 2\sec^2 \theta \tan \theta \Rightarrow dx = 2\sec^2 \theta \tan \theta d\theta$$

$$\text{When } x = 0, \theta = 0$$

$$\text{When } x = 1, \theta = \frac{\pi}{4}$$

$$\begin{aligned}\sin^{-1} \sqrt{\frac{x}{1+x}} &= \sin^{-1} \sqrt{\frac{\tan^2 \theta}{1+\tan^2 \theta}} \\ &= \sin^{-1} \sqrt{\frac{\tan^2 \theta}{\sec^2 \theta}} \\ &= \sin^{-1} \sqrt{(\sin^2 \theta)} \\ &= \sin^{-1}(\sin \theta) \\ &= \theta\end{aligned}$$

$$\int_0^1 \sin^{-1} \sqrt{\frac{x}{1+x}} dx = \int_0^{\frac{\pi}{4}} \theta \times 2\sec^2 \theta \tan \theta d\theta$$

Integrating by parts:

$$\begin{aligned}u &= \theta & u' &= 1 \\ v' &= 2\tan \theta \sec^2 \theta & v &= \tan^2 \theta\end{aligned}$$

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \theta \times 2\sec^2 \theta \tan \theta d\theta &= [\theta \tan^2 \theta]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \tan^2 \theta d\theta \\ &= \left(\frac{\pi}{4} \times 1 - 0\right) - \int \sec^2 \theta - 1 d\theta \\ &= \frac{\pi}{4} - [\tan \theta - \theta]_0^{\frac{\pi}{4}} \\ &= \frac{\pi}{4} - \left[\left(1 - \frac{\pi}{4}\right) - 0\right] \\ &= \frac{\pi}{2} - 1\end{aligned}$$

Mean mark 55%.

22. Calculus, EXT2 C1 2023 HSC 13a

$$\begin{aligned}I &= \int \frac{1-x}{\sqrt{5-4x-x^2}} dx \\ &= \frac{1}{2} \int \frac{2-2x}{\sqrt{5-4x-x^2}} dx \\ &= \frac{1}{2} \int \frac{-4-2x}{\sqrt{5-4x-x^2}} dx + \frac{1}{2} \int \frac{6}{\sqrt{5-4x-x^2}} dx\end{aligned}$$

$$\text{Let } u = 5 - 4x - x^2$$

$$\frac{du}{dx} = -4 - 2x \Rightarrow du = -4 - 2x dx$$

$$\begin{aligned}I &= \frac{1}{2} \int u^{-\frac{1}{2}} du + 3 \int \frac{1}{\sqrt{9-(x+2)^2}} dx \\ &= u^{\frac{1}{2}} + 3\sin^{-1}\left(\frac{x+2}{3}\right) + c \\ &= \sqrt{5-4x-x^2} + 3\sin^{-1}\left(\frac{x+2}{3}\right) + c\end{aligned}$$

23. Calculus, EXT2 C1 2007 HSC 1d

$$\text{Let } u = 1-x, \quad du = -dx, \quad x = 1-u$$

$$\text{When } x = 0, \quad u = 1$$

$$\text{When } x = \frac{3}{4}, \quad u = \frac{1}{4}$$

$$\begin{aligned}\int_0^{\frac{3}{4}} \frac{x dx}{\sqrt{1-x}} &= \int_1^{\frac{1}{4}} \frac{(1-u)(-du)}{\sqrt{u}} \\ &= \int_{\frac{1}{4}}^1 \left(u^{-\frac{1}{2}} - u^{\frac{1}{2}}\right) du \\ &= \left[2u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}}\right]_{\frac{1}{4}}^1 \\ &= \left[\left(2 - \frac{2}{3}\right) - \left(1 - \frac{1}{12}\right)\right] \\ &= \frac{5}{12}\end{aligned}$$

24. Calculus, EXT2 C1 2007 HSC 5c

$$\begin{aligned}
 \text{i. } (x-1)(5-x) &= 5x-x^2-5+x \\
 &= -(x^2-6x+5) \\
 &= 4-(x-3)^2 \\
 &= 2^2-(x-3)^2
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. Let } x-3 &= 2\sin\theta \\
 dx &= 2\cos\theta \, d\theta
 \end{aligned}$$

$$\text{When } x=1, \quad \theta = -\frac{\pi}{2}$$

$$\text{When } x=5, \quad \theta = \frac{\pi}{2}$$

$$\begin{aligned}
 \therefore \int_1^5 \sqrt{4-(x-3)^2} \, dx \\
 &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{(4-4\sin^2\theta)} \cdot 2\cos\theta \, d\theta \\
 &= 4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\cos^2\theta} \cdot \cos\theta \, d\theta \\
 &= 4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2\theta \, d\theta \\
 &= 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + \cos 2\theta) \, d\theta \\
 &= 2 \left[\theta + \frac{\sin 2\theta}{2} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\
 &= 2 \left[\left(\frac{\pi}{2} + 0 \right) - \left(-\frac{\pi}{2} - 0 \right) \right] \\
 &= 2\pi
 \end{aligned}$$

25. Calculus, EXT2 C1 2009 HSC 1e

$$\text{Let } x = \tan\theta, \quad dx = \sec^2\theta \, d\theta$$

$$\text{When } x=1, \quad \theta = \frac{\pi}{4}$$

$$\text{When } x=\sqrt{3}, \quad \theta = \frac{\pi}{3}$$

$$\begin{aligned}
 \int_1^{\sqrt{3}} \frac{1}{x^2\sqrt{1+x^2}} \, dx &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2\theta}{\tan^2\theta\sqrt{1+\tan^2\theta}} \, d\theta \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2\theta}{\tan^2\theta\sec\theta} \, d\theta \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec\theta}{\tan^2\theta} \, d\theta \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos^2\theta}{\cos\theta\sin^2\theta} \, d\theta \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos\theta \, d\theta}{\sin^2\theta} \\
 &= \left[-\frac{1}{\sin\theta} \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} \\
 &= \left(-\frac{2}{\sqrt{3}} + \sqrt{2} \right) \\
 &= \sqrt{2} - \frac{2\sqrt{3}}{3}
 \end{aligned}$$

26. Calculus, EXT2 C1 2010 HSC 1e

Solution 1

$$\text{Let } u = \sqrt{x}, \quad du = \frac{1}{2\sqrt{x}} dx, \quad dx = 2u du$$

$$\begin{aligned} \int \frac{dx}{1+\sqrt{x}} &= \int \frac{2u du}{1+u} \\ &= \int \left(\frac{2+2u-2}{1+u} \right) du \\ &= \int \left(2 - \frac{2}{1+u} \right) du \\ &= 2u - 2\log_e(1+u) + c_1 \\ &= 2\sqrt{x} - 2\log_e(1+\sqrt{x}) + c_1 \end{aligned}$$

Alternative Solution

$$\text{Let } u = 1 + \sqrt{x}, \quad du = \frac{1}{2\sqrt{x}} dx, \quad dx = 2(u-1) du$$

$$\begin{aligned} \int \frac{dx}{1+\sqrt{x}} &= 2 \int \frac{u-1}{u} du \\ &= 2 \int \left(1 - \frac{1}{u} \right) du \\ &= 2u - 2\log_e |u| + c_2 \\ &= 2 + 2\sqrt{x} - 2\log_e(1+\sqrt{x}) + c_2 \end{aligned}$$

NB. These solutions are equivalent by making $c_1 = c_2 + 2$

27. Calculus, EXT2 C1 2013 HSC 11d

$$\text{Let } u = 1-x^2, \quad du = -2x dx$$

$$\text{When } x = 0, \quad u = 1$$

$$\text{When } x = 1, \quad u = 0$$

$$\begin{aligned} \int_0^1 x^3 \sqrt{1-x^2} dx &= \int_0^1 x^2 \sqrt{1-x^2} x dx \\ &= \int_1^0 (1-u) \sqrt{u} \times \left(\frac{-du}{2} \right) \\ &= \frac{1}{2} \int_0^1 \left(u^{\frac{1}{2}} - u^{\frac{3}{2}} \right) du \\ &= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} - \frac{2}{5} u^{\frac{5}{2}} \right]_0^1 \\ &= \frac{1}{2} \left[\left(\frac{2}{3} - \frac{2}{5} \right) - 0 \right] \\ &= \frac{2}{15} \end{aligned}$$

STRATEGY: The choice of $u = 1 - x^2$ or $u^2 = 1 - x^2$ provided much less calculation than $u = x^2$. Take note!

28. Calculus, EXT2 C1 2013 HSC 14a

Area under curve > Area of triangle

$$\int_1^t \ln x dx > \frac{1}{2} \times (t-1) \ln t, \quad t > 1$$

$$\underbrace{\int_1^t 1 \ln x dx}_{\text{integration by parts}} > \frac{1}{2} \times (t-1) \ln t$$

$$[x \ln x]_1^t - \int_1^t x \cdot \frac{1}{x} dx > \frac{(t-1) \ln t}{2}$$

$$(t \ln t - \ln 1) - [x]_1^t > \frac{(t-1) \ln t}{2}$$

$$t \ln t - (t-1) > \frac{(t-1) \ln t}{2}$$

$$2t \ln t - 2t + 2 > t \ln t - \ln t$$

$$t \ln t + \ln t > 2(t-1)$$

$$(t+1) \ln t > 2(t-1)$$

$$\ln t > 2 \left(\frac{t-1}{t+1} \right), \quad t > 1$$

29. Calculus, EXT2 C1 2011 HSC 7b

i. Let $u = 4 - x$, $du = -dx$, $x = 4 - u$

When $x = 1$, $u = 3$

When $x = 3$, $u = 1$

$$\begin{aligned} I &= \int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)} dx \\ &= \int_3^1 \frac{\cos^2\left(\frac{\pi}{8}(4-u)\right)}{(4-u)u} (-du) \\ &= - \int_3^1 \frac{\cos^2\left(\frac{\pi}{2} - \frac{\pi}{8}u\right)}{(4-u)u} du \\ &= \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}u\right)}{u(4-u)} du \end{aligned}$$

ii. Since $\int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)} dx = \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}x\right)}{x(4-x)} dx$

We can add the integrals such that

$$\begin{aligned} 2I &= \int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)} dx + \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}x\right)}{x(4-x)} dx \\ &= \int_1^3 \frac{1}{x(4-x)} dx \end{aligned}$$

Using partial fractions:

$$\begin{aligned} \frac{1}{x(4-x)} &= \frac{A}{x} + \frac{B}{4-x} \\ 1 &= A(4-x) + Bx \end{aligned}$$

When $x = 0$, $A = \frac{1}{4}$

When $x = 4$, $B = \frac{1}{4}$

$$\begin{aligned} 2I &= \frac{1}{4} \int_1^3 \left(\frac{1}{x} + \frac{1}{4-x} \right) dx \\ &= \frac{1}{4} [\log_e x - \log_e(4-x)]_1^3 \\ &= \frac{1}{4} [\log_e 3 - \log_e 1 - (\log_e 1 - \log_e 3)] \\ &= \frac{1}{2} \log_e 3 \\ \therefore I &= \frac{1}{4} \log_e 3 \end{aligned}$$

◆ Mean mark 36%.

30. Calculus, EXT2 C1 2007 HSC 8a

i. Show $\int_0^a f(x) dx = \int_0^a f(a-x) dx$.

Let $x = a-u$, $dx = -du$

When $x = 0$, $u = a$

When $x = a$, $u = 0$

$$\begin{aligned} \therefore \int_0^a f(x) dx &= \int_a^0 f(a-u) (-du) \\ &= \int_0^a f(a-u) du \\ &= \int_0^a f(a-x) dx \quad \text{.. as required} \end{aligned}$$

ii. $f(x) = f(a) - f(a-x)$

$$\begin{aligned} \int_0^a f(x) dx &= \int_0^a [f(a) - f(a-x)] dx \\ &= \int_0^a f(a) dx - \int_0^a f(a-x) dx \\ &= \int_0^a f(a) dx - \int_0^a f(x) dx \\ 2 \int_0^a f(x) dx &= \int_0^a f(a) dx \\ &= [f(a) \times x]_0^a \\ \int_0^a f(a) dx &= \frac{f(a)}{2} (a-0) \\ &= \frac{a}{2} f(a) \end{aligned}$$

MARKER'S COMMENT: Integrating $f(a)$ was poorly done in part (ii). Pay careful attention to this in the Worked Solution.

31. Calculus, EXT2 C1 2022 SPEC2 7 MC

$$u = 1 + e^x \Rightarrow \frac{du}{dx} = e^x \Rightarrow dx = \frac{du}{e^x} = \frac{du}{u-1}$$

When $x = 0$, $u = 2$

When $x = \log_e 2$, $u = 1 + e^{\log_e 2} = 3$

$$\begin{aligned} \int_0^{\log_e 2} \frac{1}{1+e^x} dx &= \int_2^3 \left(\frac{1}{u} \cdot \frac{1}{u-1} \right) du = \int_2^{1+e^2} \left(\frac{1}{u-1} - \frac{1}{u} \right) du \\ &\Rightarrow D \end{aligned}$$

